

Yang-Jun Joo, Ph.D.

Quantitative Risk Assessment Data Scientist

Autonomous Vehicle Safety | Statistical Modeling | Large-Scale Driving Data

Orlando, FL (Open to relocation to Foster City, CA) | **Email** | Phone: +1 (407) 823-2841 | **LinkedIn** | **GitHub** | **Google Scholar**

Professional Summary: Quantitative risk assessment scientist with 8+ years of research and applied project experience developing probabilistic safety metrics, uncertainty-aware models, and automated analytics pipelines for autonomous-vehicle and transportation systems. Author of 15 published or accepted papers and PI on two competitively funded safety projects, with experience in Python, SQL, large-scale driving data, and human-behavior experiments.

TECHNICAL SKILLS

Programming & Data:	Python, SQL, R, pandas, NumPy, scikit-learn, PyTorch, Git, Linux, ETL and analytics pipelines, Apache Spark/PySpark.
Risk & Statistics:	Probabilistic risk modeling, Bayesian networks, reliability analysis, surrogate safety measures, regression and treatment-effect analysis, uncertainty quantification, conformal prediction, calibration, predictive intervals, experimental design.
Safety & Mobility:	Quantitative risk assessment, safety-performance metrics, AV safety evaluation, naturalistic driving video, connected-vehicle and telemetry data, traffic-conflict analysis, forward-collision warning evaluation, human factors.

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar	University of Central Florida	Mar 2024 – Present
- Led quantitative safety workstreams for FDOT-sponsored projects, partnering with transportation agencies and technical teams to define and validate safety-performance metrics using high-resolution traffic-controller, connected-vehicle, telemetry, and naturalistic driving data. - Built reproducible Python and SQL analytics pipelines supporting network-scale safety and mobility analyses across 1,050 arterial links and four Florida freeways, automating data cleaning, metric generation, validation, and dashboard-ready reporting. - Developed uncertainty-aware evaluation methods for autonomous-driving vision-language models using conformal prediction, identifying context-specific calibration failures in safety-critical driving-scene question answering.		
Postdoctoral Researcher	Seoul National University	Mar 2023 – Feb 2024
- Served as Principal Investigator (PI) on a KRW 73.5 million national fellowship examining how people evaluate autonomous-vehicle risk in near-miss and safety-critical scenarios. - Designed and analyzed a controlled human-subject experiment involving 926 participants, 265 driving events, and 1,325 video clips; applied regression and heterogeneous-treatment analyses to quantify the effects of vehicle automation, event severity, and participant characteristics on perceived risk.		
Graduate Research Assistant	Seoul National University	Mar 2018 – Feb 2023
- Developed probabilistic and reliability-based risk-assessment models for autonomous-vehicle lane changes, individual-driver crash risk, and proactive collision avoidance using Bayesian networks, surrogate safety measures, and risk-field methods. - Secured and led a KRW 96.1 million Global Ph.D. Fellowship and published first-author research in <i>Analytic Methods in Accident Research</i>, <i>Accident Analysis & Prevention</i>, and <i>Transportation Research Record</i>.		

SELECTED PUBLICATIONS

- Joo et al. (2023), "A Generalized Driving Risk Assessment Method for Autonomous Vehicles Using Field Theory," *Analytic Methods in Accident Research*, 37, 100251.
- Joo et al. (2022), "A Data-Driven Bayesian Network for Probabilistic Crash Risk Assessment of Individual Driver with Traffic Violation and Crash Records," *Accident Analysis & Prevention*, 174, 106721.
- Joo et al. (2026), "Context-Aware Conformal Prediction for VLM-Based Driving Scene QA," *Forty-Third International Conference on Machine Learning (ICML) Workshop EMM-QA*.

LEADERSHIP & RECOGNITION

- **Leadership & Recognition:** PI on two competitively funded projects totaling KRW 169.6M; 2023 TRB Best Paper Award (Standing Committee on Urban Transportation Data); 2021 KOTTA Best Paper Award.

EDUCATION

Seoul National University	Ph.D. in Civil & Environmental Engineering (Transportation Engineering)	Mar 2018 – Feb 2023
Seoul National University	B.S. in Civil & Environmental Engineering	Mar 2014 – Feb 2018